

Revisiting “Cross-Asset Tandem Trading and Extraordinary Volatility”: What We Changed, Why, and What We Found

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1. What this report is

We rebuilt the paper’s analysis pipeline from the ground up, re-examined its methodology end to end, and re-ran everything on a fresh 2022–2026 sample plus the March-2020 circuit-breaker days. This report explains, in plain terms: how the original paper measured things, where those methods were fragile, what we changed, why the changes are better, and what the answers look like now. Technical terms are explained the first time they appear and collected in a short glossary at the end.

The short version: the paper’s central ideas survive — the two markets really do trade in tandem, and futures really do lead — but both claims come out sharper and more *conditional* than the published versions. Futures leadership turns out to be a stress phenomenon, not a general law. Tandem trading is real and intensifies under stress, but one of the paper’s most dramatic-looking numbers was an artifact of comparing against the wrong benchmark. And the liquidity mechanism at the heart of the paper’s story is invisible at the frequency the paper measured — and unmistakable at the frequency we now measure.

2. The question and the data

The question. The S&P 500 trades in two major forms at once: SPY, an exchange-traded fund, and ES, a futures contract. They track the same index, so their prices should move together — and arbitrage traders make money keeping them together. The paper asks what happens on the wildest days: Do the two markets trade *in tandem* — buying and selling in the same direction at the same moments? Which market *sets* the price and which one follows (what economists call “price discovery”)? And does the answer change when liquidity dries up — when the standing orders that normally cushion the market get pulled?

The data. 24 full trading days, in three groups: the ten most volatile SPY sessions of 2022–2026 (the largest high-to-low intraday ranges), a calm “twin” for each of those days, and the four days in March 2020 when the market-wide circuit breaker halted all trading. Each volatile day’s twin is the same weekday roughly one year earlier — same day of the week, similar market structure, no crisis — so every “wild day” result can be compared against a matched “normal day” baseline rather than against an average.

Exhibit 1a. Volatile sessions and their paired benchmarks.

volatile session	weekday	benchmark pair	weekday	gap
2023-03-09	Thu	2022-03-24	Thu	350d
2024-07-24	Wed	2023-07-19	Wed	371d
2024-08-05	Mon	2023-08-07	Mon	364d
2024-09-03	Tue	2023-09-05	Tue	364d
2024-12-18	Wed	2023-12-20	Wed	364d
2025-01-27	Mon	2024-01-29	Mon	364d
2025-04-03	Thu	2024-04-04	Thu	364d
2025-08-01	Fri	2024-08-09	Fri	357d
2025-10-10	Fri	2024-10-18	Fri	357d
2026-06-05	Fri	2025-06-13	Fri	357d

Exhibit 1b. MWCB sessions (March 2020).

session	weekday	Level-1 halt begins	note
2020-03-09	Mon	09:34	the co-jump analysis of \$6.1
2020-03-12	Thu	09:35	
2020-03-16	Mon	09:30 (at the open)	short-selling restricted all session
2020-03-18	Wed	12:56	

Each March-2020 session contains one Level-1 market-wide circuit-breaker (MWCB) halt: the S&P 500 fell 7%, and all trading stopped for 15 minutes.

For every one of these days we have the full *order book* for both markets — not just the price, but the whole ladder of standing buy and sell orders with their sizes — snapshotted every 10 milliseconds. That is roughly 2.3 million snapshots per market per day. A one-second version of the same data is built directly from those snapshots, which matters more than it sounds (Section 5).

3. How the original paper went about it

Six pillars of the original methodology, in plain terms:

1. **Sample and handling.** Ten volatile days from 2014–2017 with paired calm days, plus the 2020 circuit-breaker days. The 15-minute trading halts on the circuit-breaker days stayed in the dataset.
2. **One-second measurement.** Prices and order flow were sampled once per second, and every estimate in the paper was computed on that one-second grid.
3. **Tandem-trading tests.** Time was cut into bars; in each bar, each market was scored as net buying or net selling. The paper counted how often the two markets pressed in the same direction and compared that count to a *coin-flip world*: each market buying or selling with 50/50 odds, independently of the other. It

then re-used the same benchmark rate (about 0.4%) when it re-cut the data at finer time scales, including “action time” (bars that advance one order at a time).

4. **What moves the correlation.** The correlation between the two markets’ returns was measured through a 100-second *rolling window* (each second, recompute the correlation of the last 100 seconds), and a statistical model (a VAR — a regression of current values on many past values) asked which order-book variables move that correlation. The number of past values (“lags”) was chosen by an automatic criterion.
5. **The liquidity mechanism.** The paper’s story — arbitrage weakens when the order book thins, letting prices drift apart — was tested on the one-second grid, largely by splitting each day into high-liquidity and low-liquidity buckets and comparing.
6. **Statistical confidence.** Significance was computed treating every second as an independent observation, and the direction-of-causality questions were resolved either by assuming an ordering (“futures move first within the second”) or by a variance-regime-shift technique (Rigobon identification).

4. The weaknesses in that approach

Each weakness below is numbered to match the pillar above.

1. **Halted markets were counted as trading.** When trading halts for 15 minutes and reopens at a very different price, a dataset that keeps the halt turns the entire gap into one giant one-second “return” — like a heart-rate monitor recording a terrifying spike because it was unplugged and plugged back in. On circuit-breaker days, that one fake observation can dominate a whole day’s statistics — in exactly the sessions the paper cares most about.
2. **One second is too slow to see who moves first.** These markets react to each other in milliseconds. Sampled once per second, nearly all of the adjustment between SPY and ES happens “within the same tick” — like photographing a hummingbird with a one-second exposure and asking which wing moved first. The standard leadership score (the Hasbrouck information share) senses this: rather than a number, it returns bounds, and at one second the bounds on a typical day are [0.004, 0.861] — a statistical shrug. Any point estimate quoted inside that range is really the *assumed ordering* talking, not the data.
3. **The coin-flip benchmark is too easy to beat.** The coin-flip world bundles two separate claims: each market is a fair coin, *and* the two are independent. On a crash day, both markets are mostly selling — each “coin” is heavily biased — so the two agree constantly even with no cross-market link at all. Rejecting the coin-flip null therefore does not demonstrate tandem trading; a one-sided day rejects it on its own. Worse, the benchmark rate is not portable across time scales: the chance that both markets are active and agree in a bar depends mechanically on how many orders land in that bar. Comparing a 48% observed rate at action time against a 0.4% per-second benchmark is comparing apples to a yardstick built for oranges.
4. **A rolling window manufactures its own dynamics.** A 100-second moving average remembers every event for exactly 100 seconds. Feed that into a model that hunts for time patterns, and it will faithfully “discover” 100-second dynamics — an echo we ourselves added, like measuring a room’s acoustics with a microphone that has its own reverb. The automatic lag-selection criterion then chases that echo (it always picks the maximum allowed), and — as we demonstrate directly in Section 6.4 — the headline

coefficients change wholesale when the lag setting changes. Results that depend on a knob setting are measuring the knob.

5. **The mechanism was tested at the wrong speed, with a fragile design.** Arbitrage capital responds to a thinning book within seconds; averaged to one-second resolution the effect is faint, and the day-by-day liquidity-bucket splits flip sign from one day to the next.
6. **The stars were too generous, and one identification tool had no grip.** Treating 23,400 seconds from the same trading day as independent observations is like polling one household a thousand times and reporting a thousand-person survey: the effective sample is 24 *days*, not millions of seconds. This is how the paper’s flagship regression got a t-statistic of 253. Separately, the Rigobon technique requires the volatile regime to change the two markets’ variances by *different proportions* — and on this sample stress scales both markets almost identically, like trying to tell which speaker is louder when someone turned up the master volume. The technique has nothing to grab.

5. The improvements, and why they are better

1. **Halt masking, everywhere.** Every snapshot inside a halt window, and the messy reopening seam after it, is excluded from every estimator — no exceptions, enforced by automated tests. *Why better:* circuit-breaker-day estimates now describe actual trading, not the unplugged monitor.
2. **Two measurement speeds, one data pull.** Everything is now measured twice: on the original one-second grid and on a 10-millisecond grid, with the one-second data *derived from the same 10ms pull*. At 10ms the markets’ reactions are genuinely spread over time, so “who moves first” becomes measurable rather than assumed — the leadership bounds tighten from [0.004, 0.861] to [0.244, 0.523] on the same day. *Why better:* the fine grid answers the question the one-second grid could only assume away; and because both grids come from one pull, any difference between them is a difference of resolution, never a difference of data. (The ES side was also rebuilt from the exchange’s own order ladder, cross-checked by independently replaying the message feed.)
3. **Two corrected benchmarks for tandem trading.** First, an independence benchmark that *keeps each market’s actual buy/sell mix* and only randomizes the link between them — so beating it demonstrates cross-market coordination, not just a one-sided day. Second, each time scale is judged against its own coincidence rate, computed from the actual number of orders per bar at that scale. *Why better:* the first isolates exactly the paper’s claim; the second makes cross-frequency comparisons meaningful.
4. **Window-free correlation measures.** The correlation dynamics are re-estimated on dependent variables that have no rolling window to echo: *RealBar* (correlation computed fresh on non-overlapping 60-second bars) as the headline, a conditional-correlation model (DCC) as corroboration, and the Hayashi–Yoshida estimator — which is immune to the sampling distortion called the Epps effect — to measure how much of the published design was measurement artifact. We also ran the old design at two lag settings (15 and 60) as a controlled experiment. *Why better:* what survives on the window-free measures is market behavior; what changes with the knob was never real.
5. **The mechanism estimated where it operates.** The arbitrage-weakening story is now estimated at 10ms, in a single pooled model across all days (with each day keeping its own baseline) in which the strength of error correction is allowed to depend continuously on the state of the order book. *Why better:* it measures

the mechanism at the speed it actually works, and replaces twenty fragile day-by-day splits with one powerful test.

6. **Honest statistics.** Confidence is now computed by clustering at the day level (24 days = 24 independent observations), by bootstrap methods built for few clusters, and by permutation tests — shuffle the “volatile”/“calm” labels across days thousands of times and ask how often chance beats the real gap. Tables with many cells get a joint correction so that one star in twenty isn’t there by luck. And instead of pretending the Rigobon technique works here, we test its precondition, report that it fails, and give the honest bracket from the two possible orderings. *Why better:* every star that survives now means something, and a referee cannot take the headline results down by attacking the error bars.

6. The results

6.1 Who sets the price: futures leadership is a stress phenomenon

Two standard scores summarize who is doing the price-setting. The *information share* (IS) asks: of the news that permanently moves the common price, what fraction shows up in ES first? The *component share* (CS) asks: when the two prices drift apart, whose price does the pair converge back toward? Both run from 0 to 1; above 0.5 means futures lead.

Exhibit 2. Full-panel price-discovery shares by sampling frequency (24 sessions).

	1s	10ms
mean IS _{ES} (midpoint of bounds)	0.484	0.542
mean CS _{ES}	0.384	0.455

How to read it: at one second, the two markets are essentially at parity (SPY slightly ahead on CS); at 10 ms, ES has a modest lead. Nowhere is there the published-era dominance.

The picture changes when the days are split by regime:

Exhibit 3. ES component share by regime.

CS _{ES}	benchmark	volatile	difference	permutation <i>p</i>
1s	0.269	0.466	+0.197	0.048
10ms	0.387	0.504	+0.117	0.047

*How to read it: on calm days SPY leads; on volatile days the futures’ share of price-setting jumps by 12–20 points, at both measurement speeds. The “permutation *p*” answers “if we shuffled the volatile/calm labels at random, how often would chance produce a gap this big?” — about 5% — so the pattern is unlikely to be luck, even with only 24 days.*

The sharpest version of this result comes from the biggest moves. At 10ms we can classify each price change as ordinary wiggle or as a *jump* (a move too large and fast for the prevailing volatility). ES’s information share is higher in the jump component than in the ordinary component (0.56 vs 0.39), and on 2020-03-09 — the first circuit-breaker day — when both markets jumped together, ES moved first 7,610 times and SPY first

1,669 times: a 4.6-to-1 ratio, with the two markets’ jumps agreeing in direction 94.8% of the time. This exhibit needs no model assumptions at all: it is simply counting who moved first.

Bottom line: the paper’s headline should change from “futures dominate price discovery” to “futures *take over* price discovery under stress.” That is a sharper claim, it is exactly the paper’s mechanism at work, and — unlike unconditional dominance — it is what the data show.

6.2 Tandem trading: real, rising with stress — and one headline number was the yardstick’s fault

Against the corrected benchmark that keeps each market’s actual buy/sell mix, coordinated trading is still clearly present, and it strengthens with stress:

Exhibit 4. Tandem order flow against the corrected independence benchmark (revised Table 5).

Panel	observed/benchmark	log-odds ratio	corner asymmetry
A. Baseline	1.27	1.098	−0.014
B. Volatile	1.34	1.385	−0.008
C. MWCB	1.37	1.470	−0.019
C’. MWCB ex-restricted day	1.36	1.413	−0.021

How to read it: “observed/benchmark” is how much more often the two markets press the same way than independence predicts, given each market’s own mix that day. The log-odds ratio is a standard strength-of-association measure — the key fact is that it climbs steadily from calm days to volatile days to circuit-breaker days. The near-zero “corner asymmetry” says buying coordination and selling coordination are equally strong — so this is genuine two-way tandem trading, not a side effect of short-selling rules.

An even more direct measurement: after stripping each market’s order flow of what is predictable from its own past, the *surprises* in SPY flow and ES flow are correlated 0.74 on calm days and 0.83 on volatile days. That is the paper’s title phenomenon measured directly — common trading pressure hitting both markets at once, rising under stress.

The correction cuts the other way at fine time scales. Judged against its own coincidence rate, each aggregation tells a different story:

Exhibit 5. Same-direction trading against the frequency-matched benchmark (revised Table 7).

Aggregation	observed	benchmark at actual counts	ratio
1 second	11.9%	0.35%	33.5
10ms	30.1%	22.9%	1.32
action time	48.4%	50.1%	0.97

How to read it: at one second, coordinated trading is 33 times its chance rate — overwhelming. At 10ms it is 1.3 times chance. At action time it is exactly at chance. The published table compared all three observed rates against the same 0.4% benchmark, which made the action-time row look like the strongest evidence in the paper; correctly benchmarked, it is no evidence at all.

Bottom line: tandem trading is a one-second-scale phenomenon — the markets coordinate over hundreds of milliseconds, not order by order. Table 5 survives its correction and gets a cleaner story (dependence rising monotonically with stress); Table 7’s cross-frequency comparison should be replaced with per-frequency ratios.

6.3 The liquidity mechanism: invisible at 1s, unmistakable at 10ms

Think of arbitrage as a rubber band keeping SPY and ES prices tied together. The paper’s mechanism says: when the order book thins, the band slackens. Our test interacts the strength of the “snap-back” (error correction) with the state of the order book, pooled across all days:

Exhibit 6. The state-dependent error-correction test across frequencies.

	1s	10ms
<i>t</i> -statistic (mid price)	0.25	5.24
<i>t</i> -statistic (microprice)	1.35	11.70
price-gap half-life on stressed books	—	~3min

How to read it: a t-statistic near zero means “no detectable effect”; above ~2 means real. At one second the mechanism is undetectable. At 10ms it is one of the strongest effects in the whole study: when books thin, the snap-back visibly weakens, and a price gap that would normally close in seconds takes about three minutes to close halfway.

Two corroborating facts. First, *cross-impact*: order flow in ES moves SPY’s price two to five times more strongly than SPY’s flow moves ES on the circuit-breaker days — the causal arrow points from futures to ETF exactly when it matters. Second, the order-book state predicts SPY’s volatility far better than the usual liquidity measure, the quoted spread (explaining 32% of the variation versus 14%) — the book-depth variables the paper introduced are the right conditioning variables.

Bottom line: the paper’s mechanism is confirmed — but only the fine grid can see it. The revision should lead with the 10ms pooled test and drop the fragile day-by-day bucket splits.

6.4 What moves the correlation between the markets — and the experiment that settles the lag question

The last system asks which order-book shocks (spreads widening, depth thinning, order-flow imbalance, volatility) drive *changes in the correlation* between the two markets’ returns. Here we ran a controlled experiment: the identical model, on identical data, with the lag setting at 15 and then at 60.

Exhibit 7. The published (rolling-window) design at two lag settings. One-unit-shock responses of the correlation measure, $\times 100$; day-clustered standard errors in parentheses; stars are jointly corrected (***/**/* = 1/5/10%).

shock	volatile, $p=15$	volatile, $p=60$	benchmark, $p=15$	benchmark, $p=60$
Spread _{ES}	-0.014** (0.004)	-0.000 (0.001)	-0.012 (0.006)	0.002 (0.001)
WtdSpread _{ES}	0.073* (0.026)	0.000 (0.001)	-0.060*** (0.011)	-0.001 (0.005)
OFI _{ES}	0.005 (0.006)	0.000 (0.002)	-0.021 (0.009)	-0.014*** (0.003)
RV _{ES}	0.342** (0.094)	0.031** (0.008)	0.517*** (0.053)	0.130 (0.053)
Spread _{SPY}	-0.028 (0.011)	0.002 (0.001)	-0.043*** (0.012)	0.002 (0.003)
WtdSpread _{SPY}	0.061*** (0.015)	0.002 (0.002)	0.092*** (0.016)	0.027** (0.007)
OFI _{SPY}	0.003 (0.003)	0.001 (0.001)	-0.007** (0.002)	0.002 (0.001)
RV _{SPY}	0.144 (0.060)	-0.018 (0.012)	0.176 (0.064)	0.056 (0.036)

How to read it: compare any cell across the $p=15$ and $p=60$ columns. Coefficients shrink up to tenfold and the pattern of stars scrambles — ten cells are starred in at least one run, and only two keep their stars in both. Nothing about the market changed between these columns; only a model setting did. This is the rolling window's echo being absorbed differently at different lag depths — the smoking gun for the weakness described in Section 4, item 4.

Exhibit 8. The window-free (RealBar) measure at the same two lag settings.

shock	volatile, $p=15$	volatile, $p=60$	benchmark, $p=15$	benchmark, $p=60$
Spread _{ES}	0.372 (0.159)	1.521 (0.630)	-0.313*** (0.057)	-0.578 (0.502)
WtdSpread _{ES}	0.960*** (0.160)	5.412*** (0.729)	-0.077 (0.107)	0.456 (0.617)
OFI _{ES}	-0.083 (0.090)	0.711 (0.419)	-0.317** (0.078)	-1.538*** (0.370)
RV _{ES}	0.281 (0.167)	6.106*** (1.013)	1.604*** (0.322)	11.750*** (0.808)
Spread _{SPY}	-0.010 (0.251)	0.984 (0.683)	0.157 (0.237)	1.051 (0.754)
WtdSpread _{SPY}	0.329 (0.139)	0.810 (0.776)	0.393* (0.138)	1.255 (0.452)
OFI _{SPY}	0.033 (0.103)	1.249*** (0.309)	-0.010 (0.091)	-0.387 (0.423)
RV _{SPY}	0.717** (0.205)	1.973 (0.738)	0.210 (0.561)	1.480 (0.569)

How to read it: sizes still move with the lag setting (coefficient sizes in this kind of model are denominated in model-dependent units, at either lag depth), but the inference is stable: the same three cells are significant at both settings, no significant cell changes sign, and no cell is significant with opposite signs in the two runs.

What survives everywhere — across both lag settings and every measurement design: volatility shocks raise the markets' correlation (ES volatility is the one shock significant in every design at both settings); order-book-depth shocks matter, with a sign that depends on regime (an ES-side book-thinning shock raises correlation on volatile days and lowers it on calm days); ES order-flow imbalance lowers calm-day correlation; and the microprice deviation variable does nothing anywhere.

Bottom line: the published Table 9’s coefficient magnitudes were, to first order, echoes of the measurement window. The revision should rebuild the table on the window-free measure, present it as a signs-and-significance exhibit, fix the lag setting in advance, and move the published-design comparison to an appendix as the demonstration of the artifact.

6.5 What the honesty upgrades did to the stars

The flagship regression’s t-statistic falls from 253 (treating every second as independent) to 4.6 with day-clustering — still decisively significant ($p = 0.001$), now honestly so. The regime differences in Exhibit 3 survive the permutation test at both frequencies. And the Rigobon identification technique, tested rather than assumed, turns out to have no grip on this sample (the stress regime scales both markets’ variances nearly equally) — so the revision should report the assumption-based bracket and the failed precondition, not het-ID point estimates. Every result quoted in this report carries the corrected inference.

7. What the paper should now say — recommendations

1. **Recast the headline:** “futures leadership is a stress phenomenon.” Near parity on average; a 12–20-point futures takeover in the volatile regime (chance probability ~5%); largest at the discontinuities (jump share 0.56; co-jump lead 4.6:1).
2. **Keep Table 5** on the corrected independence benchmark (dependence rising monotonically with stress); **replace Table 7’s** cross-frequency comparison with per-frequency ratios, and let the action-time row say what it now says: at the order-by-order scale, coordination is at chance.
3. **Quote leadership shares at 10ms** (where they are measurements, not assumptions), with 1s as the robustness column.
4. **Lead the mechanism section with the pooled 10ms error-correction test** ($t = 5.2$ and 11.7 ; three-minute stressed half-lives) and the cross-impact asymmetry; drop the day-by-day liquidity-bucket splits.
5. **Adopt the honest inference everywhere:** day-clustering, few-cluster bootstrap, permutation tests as the headline evidence at 24 days, joint corrections on multi-cell tables; report the Rigobon precondition failure and the ordering bracket instead of het-ID estimates.
6. **Add the innovation-level tandem correlation** (0.74 calm $\rightarrow 0.83$ volatile) as the direct measurement of the title phenomenon.
7. **Rebuild Table 9 window-free** (RealBar headline, DCC corroboration), signs and significance only, lag setting fixed in advance; move the rolling-window-vs-HY contrast to the appendix as the measurement-artifact demonstration.

8. Caveats and open items

- **Futures contract rolls.** On three sessions the calendar rule for choosing the futures contract month is debatable: 2020-03-18 and 2024-12-18 land mid-roll (the chosen contract carries 69.5% and 64.2% of volume), and on 2025-06-13 the rule picked the minority contract (13.3% of volume). We must either re-extract that day pinned to the majority contract and say so, or keep the rule and report the share — it cannot pass silently.

- **2020-03-16** had short-selling restricted all session; every pooled result is reported with and without it.
- **Quote staleness at 10ms.** At that speed most snapshots show no price change (80% for SPY, 89% for ES); all fine-grid second-moment estimates use noise-robust methods, and correlation at the fine grid is always computed on bars, never tick by tick.
- **One open validation item.** Our independent replay of the ES message feed disagrees with the vendor's order ladder on one day (2024-12-18); until that is diagnosed, the ladder validation exhibit stays out of the draft.

Appendix: glossary

SPY / ES. The S&P 500 ETF (trades on stock exchanges) and the E-mini S&P 500 futures contract (trades on CME). Same index, two markets.

Order book / ladder. The standing list of buy orders (bids) and sell orders (asks) at each price, with sizes. “Depth” is how much size is resting there.

Grid / frame. A frame is one snapshot of both order books. The fine grid snapshots every 10ms; the coarse grid (1s) is built from the same snapshots.

Mid / microprice. Mid = halfway between best bid and best ask. The microprice is a depth-weighted version that leans toward where the book imbalance says price is headed.

Spread / WtdSpread. The quoted spread is the gap between best bid and best ask. The weighted spread is the *effective* round-trip cost of actually executing a realistic size, walking down the ladder — it sees book thinning that the quoted spread misses.

OFI (order-flow imbalance). Net buying-vs-selling pressure read off order-book changes: depth arriving on the bid side counts positive, on the ask side negative.

RV (realized variance). A running measure of how violently the price has been moving.

Price discovery. Which market impounds new information into the price first.

IS / CS (information share / component share). The two standard 0-to-1 scores of price discovery leadership; above 0.5 means the futures lead. IS is reported as bounds because it depends on an ordering assumption; CS does not.

Error correction / VECM. The statistical model of two prices tied together: when they drift apart, the model measures how fast each snaps back. The “state-dependent” version (ECM-SDE) lets the snap-back strength depend on the order book's condition.

VAR / lags. A regression of current values on past values; the “lag” setting is how far back the model looks.

Rolling window. Recomputing a statistic each period over the last W periods; smooths, but also smears every event over exactly W periods.

RealBar. Correlation computed fresh on non-overlapping 60-second bars — no rolling window, so no smearing.

Hayashi–Yoshida (HY). A correlation estimator built for asynchronous data; immune to the *Epps effect*, the mechanical fade of measured correlation at very fine sampling.

DCC. A model-based conditional correlation (GARCH family); window-free corroboration.

Lee–Mykland. A statistical test that classifies each price move as ordinary or as a jump, relative to prevailing local volatility.

Rigobon identification. A technique that uses volatility-regime shifts to establish causal direction; requires the regimes to change the two markets’ variances by different proportions (testable — fails here).

Cholesky ordering / bracket. The assumption that one market moves first within the sampling interval; computing results under both orderings gives an honest bracket.

Day-clustering / wild-cluster bootstrap / Webb weights. Ways of computing error bars that treat each *day* (not each second) as one independent observation, built to work with as few as 4–24 clusters.

Permutation test. Shuffle the group labels (volatile/calm) across days thousands of times; the p-value is the fraction of shuffles that beat the real gap.

Romano–Wolf correction. Adjusts significance stars when a table has many cells, so a few stars can’t appear by luck alone.

Action time. Bars that advance one order arrival at a time instead of by the clock.

MWCB. Market-wide circuit breaker: at a 7% S&P 500 decline, all trading halts for 15 minutes.

GFEVD. A variance decomposition that respects the measured correlation between shocks instead of assuming them uncorrelated.

Open items: the ESH5 2024-12-18 validation diagnosis, and the 2025-06-13 contract-month decision.